NATIONAL INSTITUTE OF TECHNOLOGY, TIRUCHIRAPPALLI - 620015

Instrumentation and Control Engineering

CEIR11 – BASICS OF CIVIL ENGINEERING

Cycle Test 1

Semester: I
Batch: ICE B
Date: 13-10-2026

Session: July 2026
Maximum Marks: 20
Time: 4.30 PM – 5.30 PM

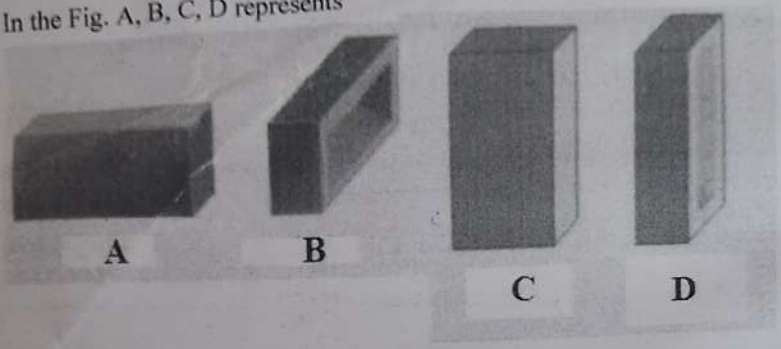
PART – A

(Each question in Part-A carries 1 mark each)

1. What is the usual range of slump for concrete used in normal reinforced cement concrete (RCC) works?
   a) 0–10 mm b) 25–50 mm c) 75–100 mm d) 150–200 mm
2. Three concrete cubes of size 150 x 150 x150 mm were tested for compressive strength. The failure loads were 720 kN, 765 kN, and 690 kN, respectively. Calculate the average compressive strength of the concrete.
   a) 30.0 N/mm² b) 32.2 N/mm² c) 32.5 N/mm² d) 33.0 N/mm²
3. The ratio of different ingredients (cement, sand and aggregate) in a concrete mix of grade M 20 is.....
4. Which of the following pairs is incorrectly matched?

   a) Granite – Igneous rock
   b) Sandstone – Sedimentary rock
   c) Slate – Metamorphic rock
   d) Basalt – Sedimentary rock
5. Which of the following combinations correctly represents the desirable composition of silica and alumina in good brick earth?
   a) Silica: 20–30%; Alumina: 50–60%
   b) Silica: 50–60%; Alumina: 20–30%
   c) Silica: 5–6%; Alumina: 20–30%
   d) Silica: 50–60%; Alumina: 5–6%
6. A concrete mix has a water–cement ratio of 0.50. If the cement content is 360 kg/m³, how much water is required per cubic metre?
   a) 160 litres b) 180 litres c) 200 litres d) 220 litres
7. A cement sample shows an expansion of 14 mm in the Le-Chatelier test. Based on the stated limit, the cement should be classified as:
   a) Sound b) Unsound c) Rapid-hardening d) Quick-setting
8. Which constituent imparts plasticity to brick earth but may cause excessive shrinkage and warping when present in excess?
   a) Silica b) Lime c) Alumina d) Magnesia

9. In the Fig. A, B, C, D represents



a) A- Shiner, B- Rowlock, C-Soldier, D-Sailor
b) A-Rowlock, B-Shiner, C-Soldier, D-Sailor
c) A- Shiner, B- Rowlock, C-Sailor, D-Soldier
d) A-Soldier, B-Shiner, C-Rowlock, D-Sailor

10. A brick weighs 2.0 kg when dry and 2.4 kg after immersion in water. What is its water absorption by weight?
    a) 10% b) 15% c) 20% d) 25%
11. The percentage of tricalcium aluminate (C_3A) in cement typically ranges from:
    a) 5–11% b) 9–14% c) 21–45% d) 25–50%

PART – B

(Each question in Part-B carries 3 mark each)

12. Explain briefly on **a)** Shotcrete **b)** Refractory Concrete **c)** High Alumina Cement
13. Describe in detail the various operations involved brick manufacturing process.
14. *Explain* the various tests to determine the mechanical properties of stones.